

Technology Stack

Date	30 September 2026
Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client	HTML5, CSS3, Jinja2 templates	Simple responsive web UI and server-rendered pages.
2	Backend / Server	Python, FastAPI, Uvicorn	Fast API development, routing, form handling and lightweight deployment.
3	Database / Storage	SQLite, SQLAlchemy	Lightweight relational persistence suitable for a student prototype.
4	AI / GenAI	Google Gemini API, gemini-3.5-flash-lite	Generates personalized workouts, nutrition tips and revised plans.
5	Authentication / Sessions	SessionMiddleware + environment credentials	Protects admin dashboard access without hard-coding credentials.
6	Version Control	Git / GitHub	Source control, collaboration and project backup.
7	Environment / Configuration	python-dotenv	Loads API keys and session/admin secrets from .env.